ITM SKILLS UNIVERSITY

SCHOOL OF FUTURE TECH

Case Study 186 : C++

Case Study Topic : Online Survey Analysis System

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Index ~

1/ Introduction to the Case Study

2/ Problem Statement / Case Background (Abstract)

3/ Problem Statement / Case Study Design

4/ Methods & Algorithms Technology Applied in the Problem Statement / Case Study

5/ Problem Statement / Case Study Implementation Details and Snapshots

6/ Problem Statement / Case Study Results and Conclusion

7/ References

Introduction to the Case Study ~

Surveys are widely used to collect feedback from users in areas such as education, business, customer service, and product evaluation. Analyzing survey responses helps organizations understand user satisfaction and make informed decisions.

This case study focuses on developing an Online Survey Analysis System using C++. The system allows users to submit ratings for a survey and analyzes the collected data to generate useful statistics such as the average rating and total number of responses.

The program demonstrates how basic programming concepts like arrays, loops, conditional statements, and user input can be used to create a simple yet effective data analysis tool. This system provides a menu-driven interface where users can submit responses and view survey analysis results.

Problem Statement / Case Background (Abstract) ~

Background -

Organizations and institutions frequently conduct surveys to gather opinions and feedback from users. However, manually analyzing survey responses can be time-consuming and inefficient. A simple automated system can help store responses and perform basic statistical analysis quickly.

Abstract -

This case study presents the design and implementation of an Online Survey Analysis System using C++. The system allows users to enter numeric ratings between 1 and 5 for a survey. These responses are stored in an array and analyzed to calculate the average rating and total responses.

The program uses a menu-driven structure, allowing users to :

- Submit survey responses
- View survey analysis
- Exit the program

The system demonstrates how basic programming techniques can be applied to process survey data and generate summary statistics efficiently.

Case Study Design ~

The Online Survey Analysis System is designed with a simple interactive structure consisting of the following components :

Survey Input -

Users can submit a rating between 1 and 5 representing their opinion or satisfaction level.

Data Storage -

All ratings are stored in an array that can hold up to 100 survey responses.

Data Analysis -

The system processes the stored ratings and calculates :

- Average rating
- Total number of responses

Menu-Driven Interface -

The program provides the following menu options :

1/ Submit Response

2/ View Analysis

3/ Exit

Output Display -

The results of the analysis are displayed to the user in a clear summary format.

Methods & Algorithms Technology Applied in the Case Study ~

The following methods and techniques are used in the system.

Data Collection -

- Users enter ratings through the console.
- The rating must be between 1 and 5.

Data Storage -

- Ratings are stored using an array (float r[100]).
- The array allows storing up to 100 responses.

Data Processing -

Basic statistical calculations are used :

Average Rating Formula 👇

Average Rating = Sum of all ratings / Number of responses

This helps measure overall user satisfaction.

Input Validation -

The program checks whether the rating entered is within the valid range.

Example condition used :

rating >= 1 && rating <= 5

If the rating is invalid, the system shows an error message.

Program Control Structure -

The program uses :

- Do-While Loop – to repeatedly display the menu
- If-Else Conditions – to perform actions based on user choice
- Arrays – to store survey responses

Case Study Implementation Details and Snapshots ~

Implementation Overview -

The system is implemented as a C++ console-based application. It consists of a single program file that handles :

- User input
- Response storage
- Data analysis
- Result display

Key Implementation Steps -

1/ Display the survey menu.

2/ Allow users to submit ratings.

3/ Store ratings in an array.

4/ Calculate average rating.

5/ Display analysis results.

6/ Allow the user to exit the program.

Example Menu Interface :

```
--- Online Survey Analysis System ---
1/ Submit Response
2/ View Analysis
3/ Exit
```

Example Input :

```
⭐ Enter rating (1 - 5): 4
✅ Response submitted successfully!
```

```
⭐ Enter rating (1 - 5): 4
✅ Response submitted successfully!
```

Example Output :

```
📈 --- Survey Analysis --- 📈
Average Rating: 4
Total Responses: 2
```

Program Code -

```
// Online Survey Analysis System



#include <iostream>

using namespace std;



int main()

{

   // Variables to store rating, total sum, and number of responses

   float rating, sum = 0;



   // Array to store up to 100 survey responses

   float r[100];



   int count = 0;   // Counts number of responses

   int choice;      // Stores menu choice
```

```
do
{
    // Display survey menu
    cout << "\n--- Online Survey Analysis System ---\n";
    cout << "1/ Submit Response\n";
    cout << "2/ View Analysis\n";
    cout << "3/ Exit\n";
    cout << "👉 Enter your choice: ";
    cin >> choice;


    // Option 1: Submit a new survey response
    if (choice == 1)
    {
        cout << "⭐ Enter rating (1 - 5): ";
        cin >> rating;
```

```
        // Check if rating is valid
        if (rating >= 1 && rating <= 5)
        {
            r[count] = rating;   // Store rating in array
            sum += rating;       // Add rating to total sum
            count++;             // Increase response count

            cout << "✅ Response submitted successfully!\n";
        }
        else
        {
            cout << "❌ Invalid rating! Please enter between 1 and 5.\n";
        }
    }

    // Option 2: View survey analysis
    else if (choice == 2)
```

```
    {
        // If no responses yet
        if (count == 0)
        {
            cout << "⚠️ No responses yet.\n";
        }
        else
        {
            // Calculate average rating
            float avg = sum / count;

            cout << "\n📈 --- Survey Analysis --- 📈\n";
            cout << "Average Rating: " << avg << endl;
            cout << "Total Responses: " << count << endl;
        }
    }
```

```
    }

    while (choice != 3);   // Loop continues until user selects Exit



    // Exit message

    cout << "🙏 Thank you for using the Survey System!\n";



    return 0;

}
```

Output -

```
● sanikagorakhnathkangane@Sanikas-MacBook-Air C++ Case Study (OSAS) % clang++ osas.cpp -o osas
● sanikagorakhnathkangane@Sanikas-MacBook-Air C++ Case Study (OSAS) % ./osas

  --- Online Survey Analysis System ---
  1/ Submit Response
  2/ View Analysis
  3/ Exit
  👉 Enter your choice: 1
  ⭐ Enter rating (1 - 5): 4
  ✅ Response submitted successfully!

  --- Online Survey Analysis System ---
  1/ Submit Response
  2/ View Analysis
  3/ Exit
  👉 Enter your choice: 1
  ⭐ Enter rating (1 - 5): 4
  ✅ Response submitted successfully!

  --- Online Survey Analysis System ---
  1/ Submit Response
  2/ View Analysis
  3/ Exit
  👉 Enter your choice: 2

  📈 --- Survey Analysis --- 📈
  Average Rating: 4
  Total Responses: 2

  --- Online Survey Analysis System ---
  1/ Submit Response
  2/ View Analysis
  3/ Exit
  👉 Enter your choice: 3
  🙏 Thank you for using the Survey System!
○ sanikagorakhnathkangane@Sanikas-MacBook-Air C++ Case Study (OSAS) %
```

Case Study Results and Conclusion ~

Findings -

The implemented system successfully demonstrates how survey responses can be collected and analyzed using basic programming techniques.

Key observations :

- The program efficiently stores multiple responses.
- Average rating calculation provides quick insight into user satisfaction.
- Input validation ensures that only valid ratings are accepted.

Conclusion -

The Online Survey Analysis System provides a simple and effective way to collect and analyze survey data. The system demonstrates how fundamental C++ programming concepts such as arrays, loops, and conditional statements can be applied to real-world problems.

Although the system is simple, it forms a foundation for more advanced survey systems that could include features like graphical interfaces, databases, or online survey forms.

References ~

1/ C++ Programming Documentation – Standard Input/Output Library (iostream)

2/ Basic Programming Concepts in C++

3/ Tutorials on Arrays and Loops in C++

4/ Academic resources on Survey Data Analysis

5/ Online educational materials related to C++ programming

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https://www.geeksforgeeks.org/c-plus-plus/

W3Schools 🎓

https://www.w3schools.com/cpp/

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